

Effective Communication

Shannon E. Ellis, Ph.D
UC San Diego



Department of Cognitive Science
sellis@ucsd.edu

Written Communication



Data Science Reports

1. In-depth details of analysis
2. Full Explanation (nothing extra)
3. A handful of figures (w/ interpretation)
4. Tell a Story



What to avoid:

1. Explaining every little thing you did
2. Being too wordy
3. Using the wrong visualization/bad figures
4. Typos & Confusion

What to Include in COGS 108 Final Project Report

- **Good Title** - short and informative
- **Abstract** - *brief* overview w/ results
- **Background** (+ references) - intro to topic + prior work
- **Hypothesis** - informed guess + why
- **Datasets** - what data you used
- **Data Cleaning** - what you did to clean/check the data
- **Data Analysis** - explore the data, explain the process, answer the ?
- **Ethics & Privacy** - consider the entire process
- **Conclusion & Discussion** - wrap it up & discuss limitations

Final Project: Video

3% of Final Grade

3-5 minutes

All members must be involved but it's not required that all members speak or that members' faces are on video.

Can be a slideshow presentation w/ voiceover.

Can be something more creative. Has to effectively communicate your project.

Oral Communication

01:

Your Audience



02:

Storytelling



03:

The Grammar of Graphics



04:

The Glamour of Graphics





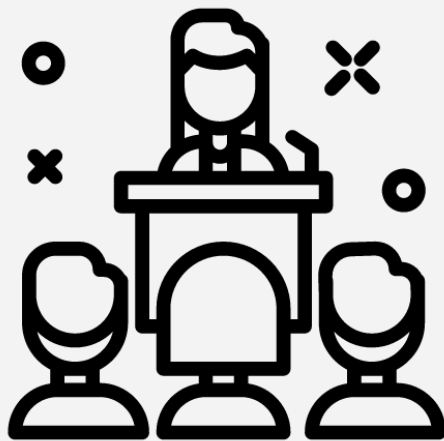
Presentation Goals

1. Inspire
2. Educate
3. Entertain



Ground Rules

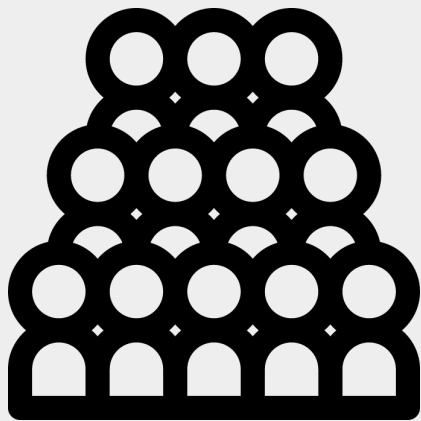
1. Prepare
2. Practice
3. Don't go over time



01:

Your Audience

Your Audience



Consider your audience.

- General vs. technical?
 - Audience background?
 - Setting?
-

General

✓ background

⊘ limit technical details

🎉 emphasize take-home

Presentation: General Audience

Introduction & Background

- Details on background material
- Full explanation of question and why doing analysis



Analysis

- Limited discussion of approach
- Focused presentation of results



Conclusion

Your Audience

General

✓ background

🚫 limit technical details

🎉 emphasize take-home

Technical

⬇️ limit background

💻 all-the-details

🎉 emphasize take-home

Presentation: Technical Audience

Introduction
&
Background



Analysis

- Details on methodology
- Detailed results
- Discussion of tools/approach

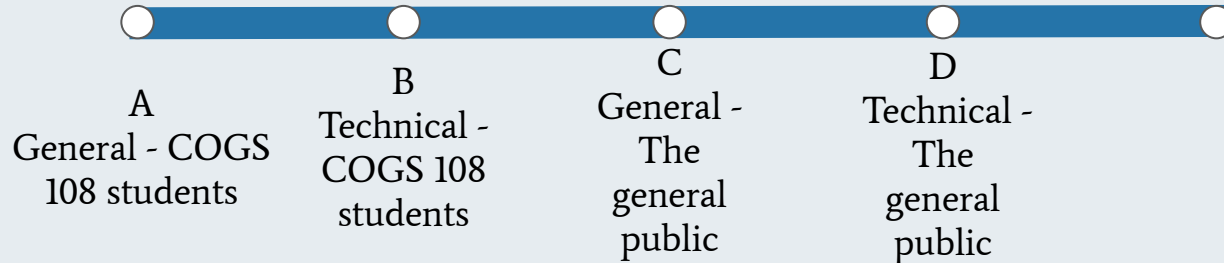
Conclusion



Final Project Video



Q1: Who's your audience for the Final Project Video?





02:

Storytelling



Storytelling: Ground Rules

1. Enticing, short title
2. Clear presentation
3. All the necessary info
4. Nothing extra



Outline Your Talk

1. Introduction
2. Topics (1, 2, 3)
3. Conclusion

In other words...
repetition is OK



Outline Your Talk

1. Tell them what you'll tell them.
2. Tell them.
3. Tell them what you told them.

Checklist

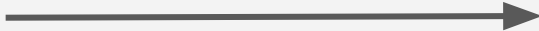
Talk

- ☐ Has a good title
- ☐ Tells a story
- ☐ Only essential info on slides
- ☐ Appropriate level of detail for audience
- ☐ Practiced
- ☐ Orienting audience to all figures
- ☐ Within time limit
- ☐ Take-home message explicitly clear

On your slides...

- Limit number of ideas
- Limit words
- Choose good fonts
- Make text readable
- Include references

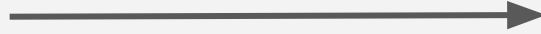
Slide Design Matters



This is not very easy to read.

This is much
easier to read.

Use Large (enough) Text



*This is not very
easy to read.*

This is much
easier to read.

Choose Good Fonts

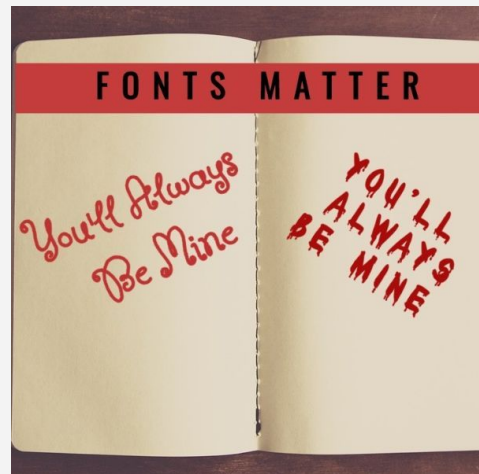
Fonts matter



MEGAFLICKS



Fast Taco



Fonts matter

Roboto family

Source family

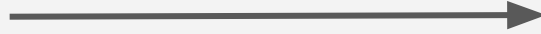
Fira family

Alegreya

Crete Round

Muli

<https://practicaltypography.com/>



This is not
very easy to
read.

This is much
easier to read.

Choose Good Colors

Color is Hard.

- Figures
 - Slides
 - Highlighting
-

Rule #1: Choose intuitive colors.

NOT IDEAL



GOOD
BAD



FOREST
LAKE



FEMALE
MALE

BETTER



GOOD
BAD



FOREST
LAKE



FEMALE
MALE

Rule #2: Consistency is key.

NOT IDEAL

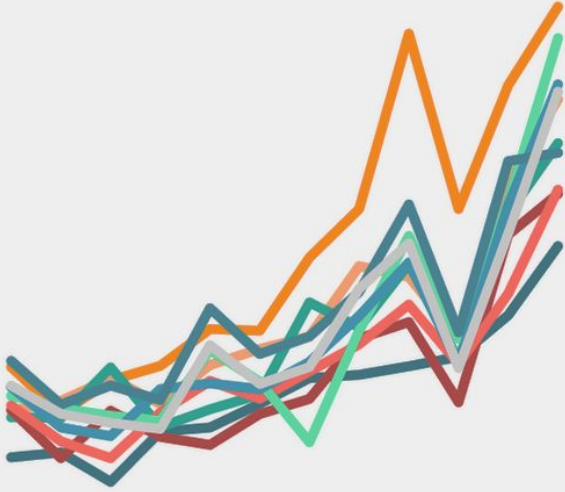


BETTER

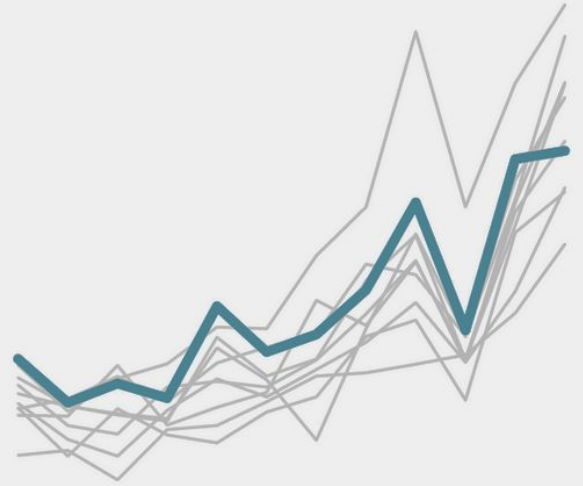


Rule #3: Gray is (definitely) your friend.

NOT IDEAL



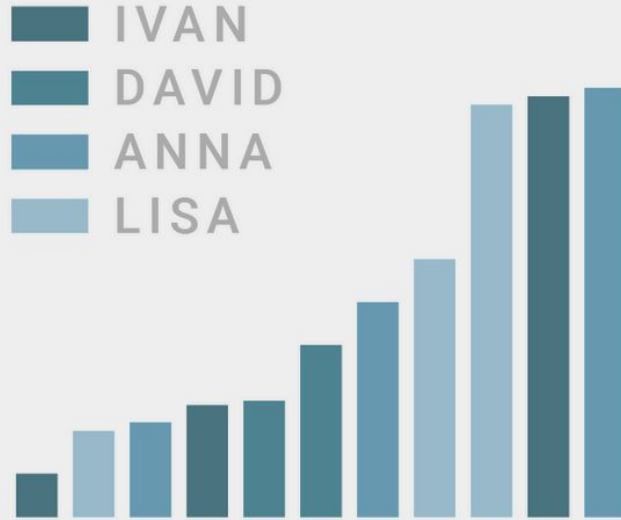
BETTER



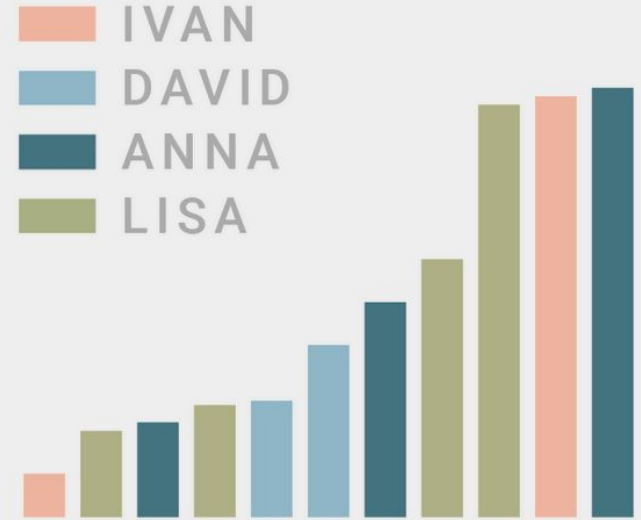
<https://blog.datawrapper.de/colors/>

Bonus: Gradients for continuous. Distinct for categories

NOT IDEAL

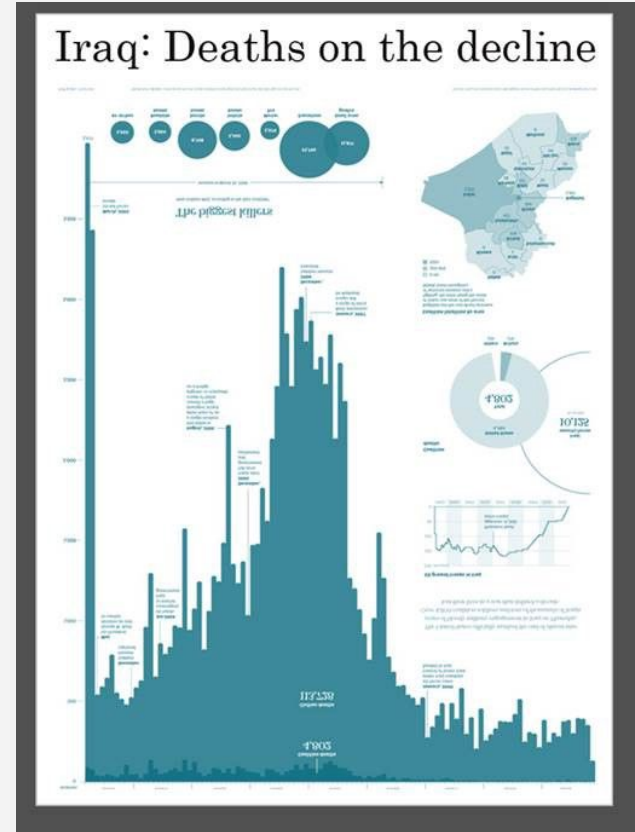


BETTER

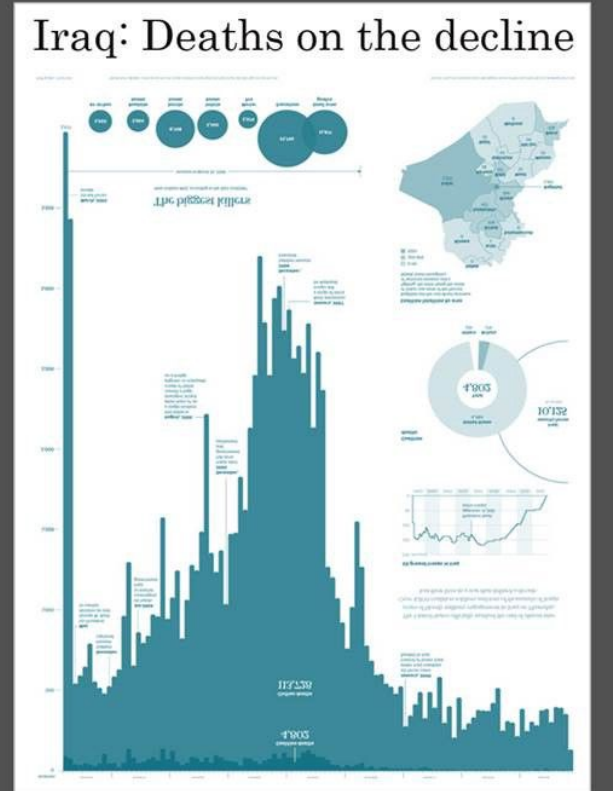


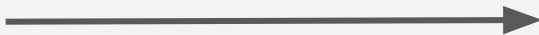
<https://blog.datawrapper.de/colors/>

Design choices alter the tone of your message



https://medium.com/@Elijah_Meeks/what-charts-say-6e31cbba2047





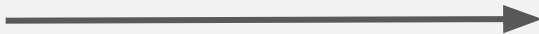
Title

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Size

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



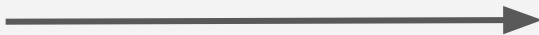
Title

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Weight

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



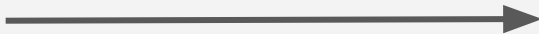
Title

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Color

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



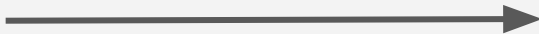
Title

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Spacing

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



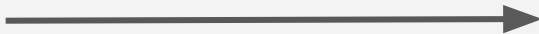
Title

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Typeface

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



Title

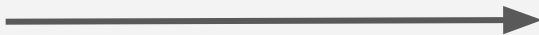
This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Title

SUBTITLE

This text is not really important, and it's not where I want you to focus your attention, so I'm going to play around with some things that allow you to focus on what I want you to focus on.

Hierarchy for Attention



- You waste others' time
- IF YOU
DON'T
ALIGN
STUFF
- It's distracting
 - It's unhelpful to your viewers

ALIGN STUFF

- Worth the time
- Doesn't distract
- Helps viewers

Take the Time to Align

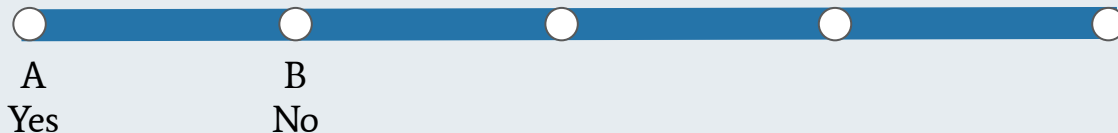
Slides

- ☐ Consistent theme throughout
- ☐ Tone sets appropriate theme
- ☐ Good, consistent fonts
- ☐ Consistent, accessible colors
- ☐ Important stuff highlighted
- ☐ Alignment looks good
- ☐ Appropriate text/font size
- ☐ Appropriate whitespace
- ☐ Colors will project well

Final Project Video



Q2: Should you scroll through your jupyter notebook during your final project presentation?



Visual Communication



Good Figures: Ground Rules

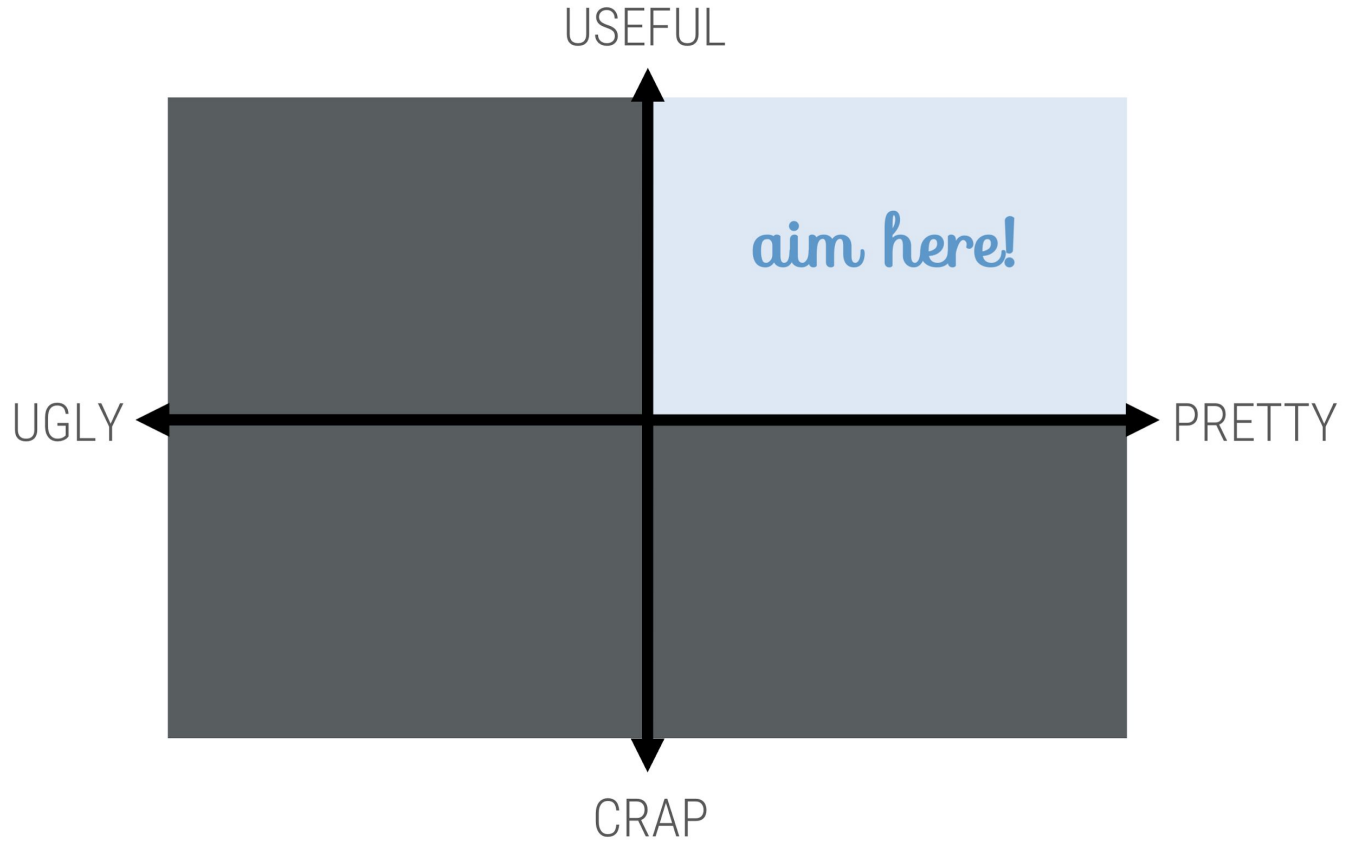
1. Left-align titles at top-left
2. Don't make people tilt their head
3. Alignment should create clean lines & symmetry
4. ~~Borders~~ - don't use them
5. Remove and lighten as much as possible
6. Legends suck
7. White space is like garlic - take the amount you need and triple it

NOTE! These are directly taken from Will Chase's rstudio::conf2020 talk:
<https://www.williamrchase.com/slides/assets/player/KeystoneDHTMLPlayer.html>

Take a sad plot & make it better

Inspiration: Allison Hill & <https://policyviz.com/helpmeviz/>

Iteration



Source: Jackie Wirz

Prevalence and access to services

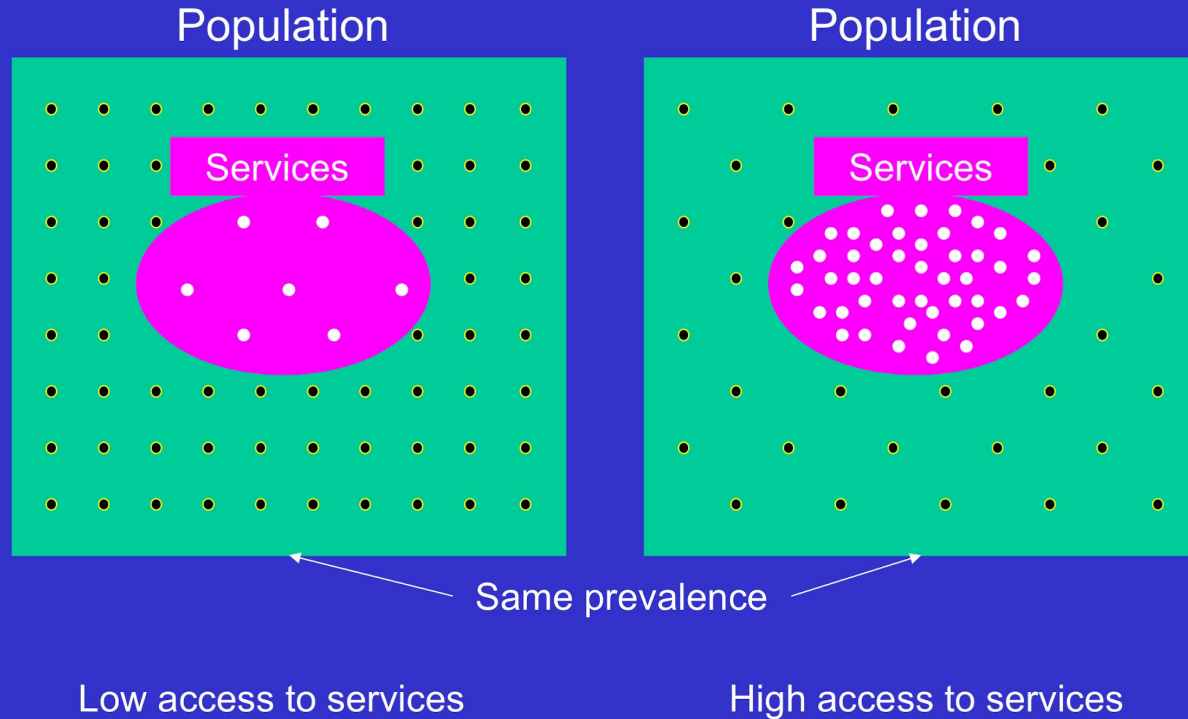
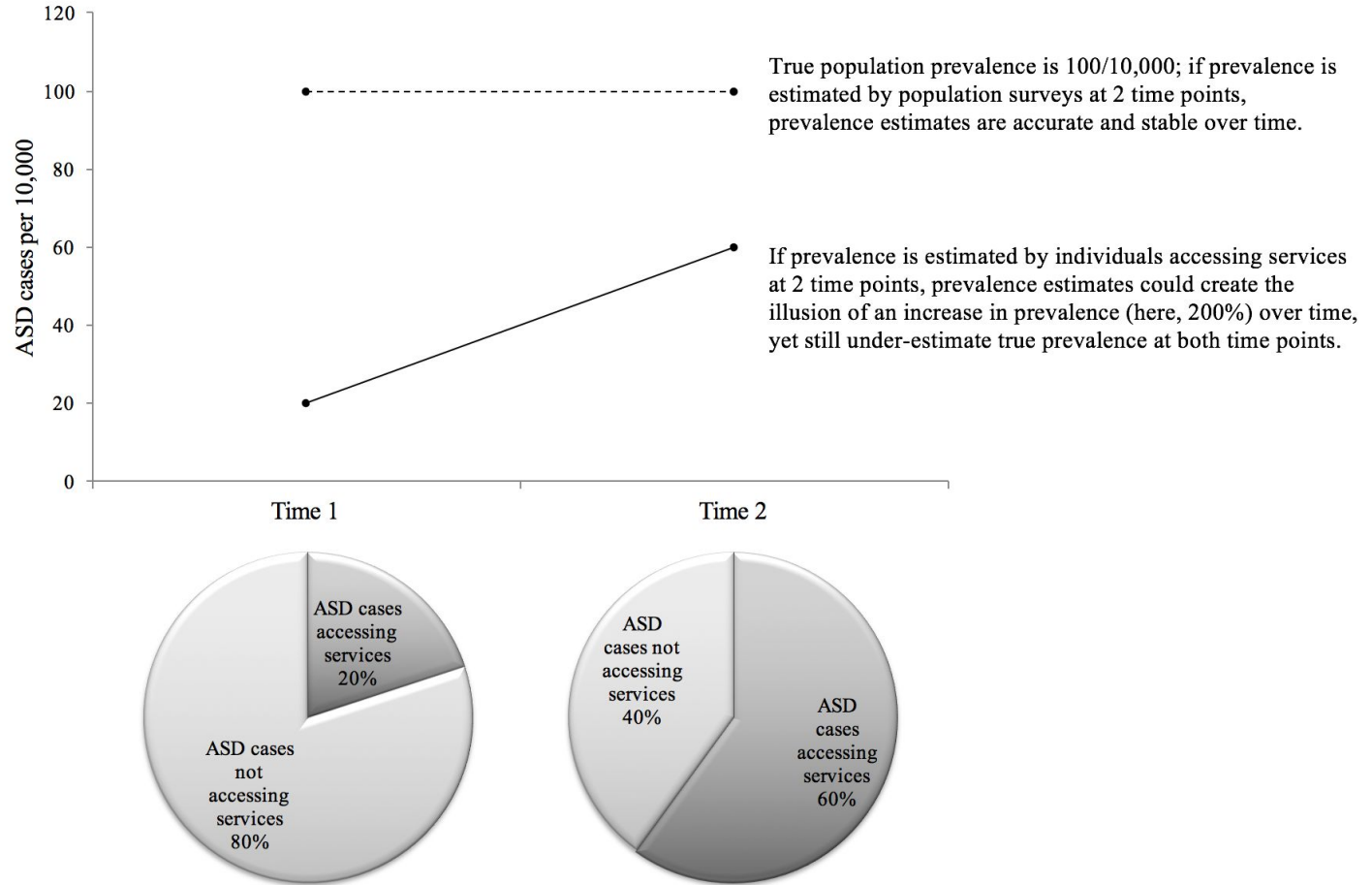
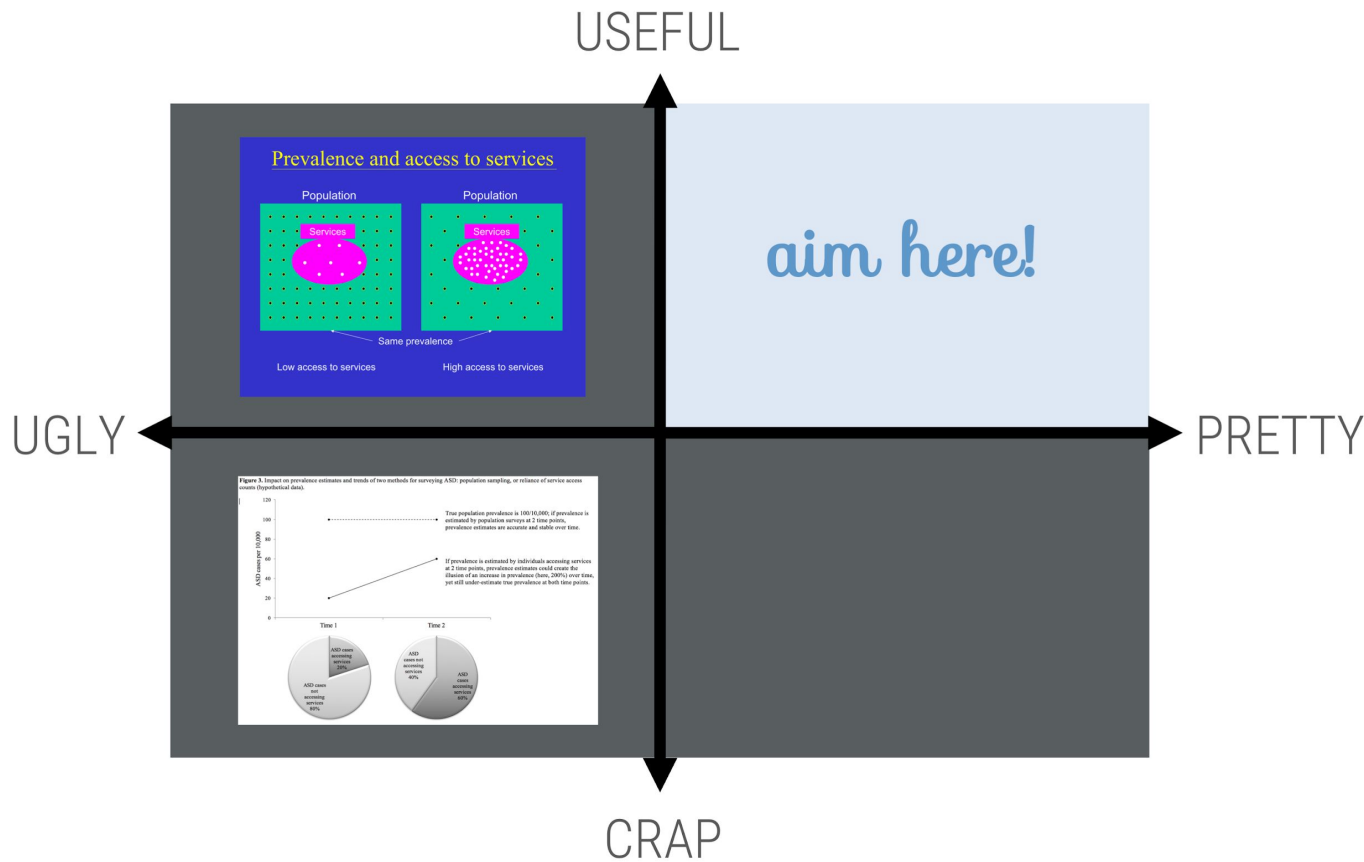


Figure 3. Impact on prevalence estimates and trends of two methods for surveying ASD: population sampling, or reliance of service access counts (hypothetical data).

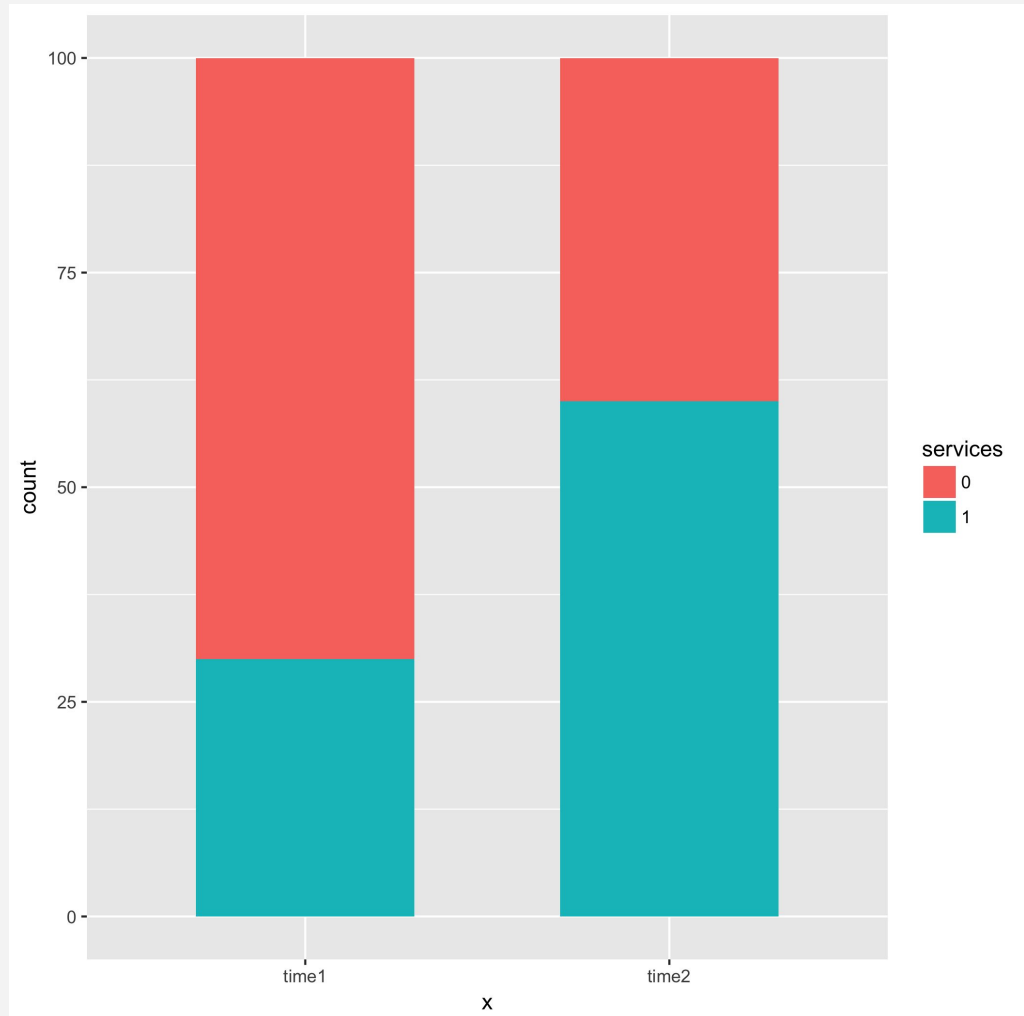


Iteration



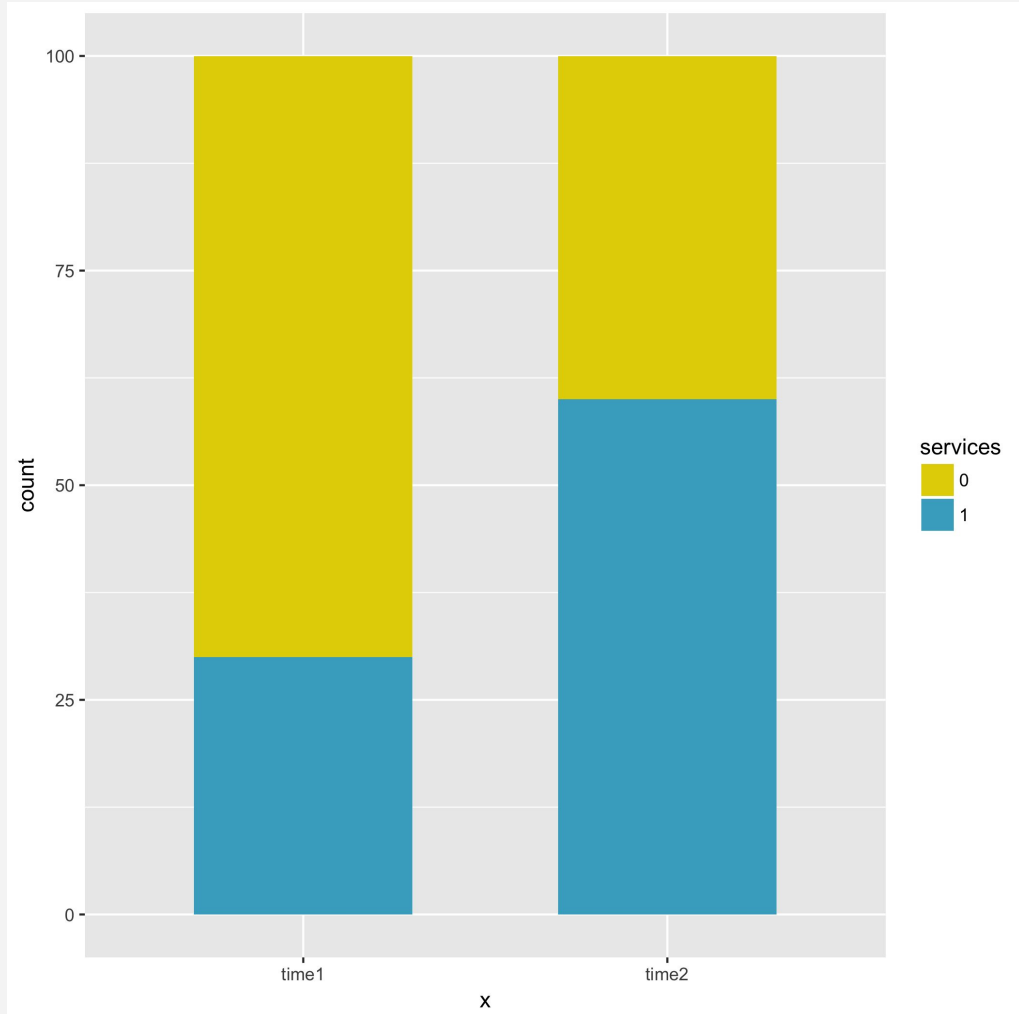
Source: Jackie Wirz

Iteration



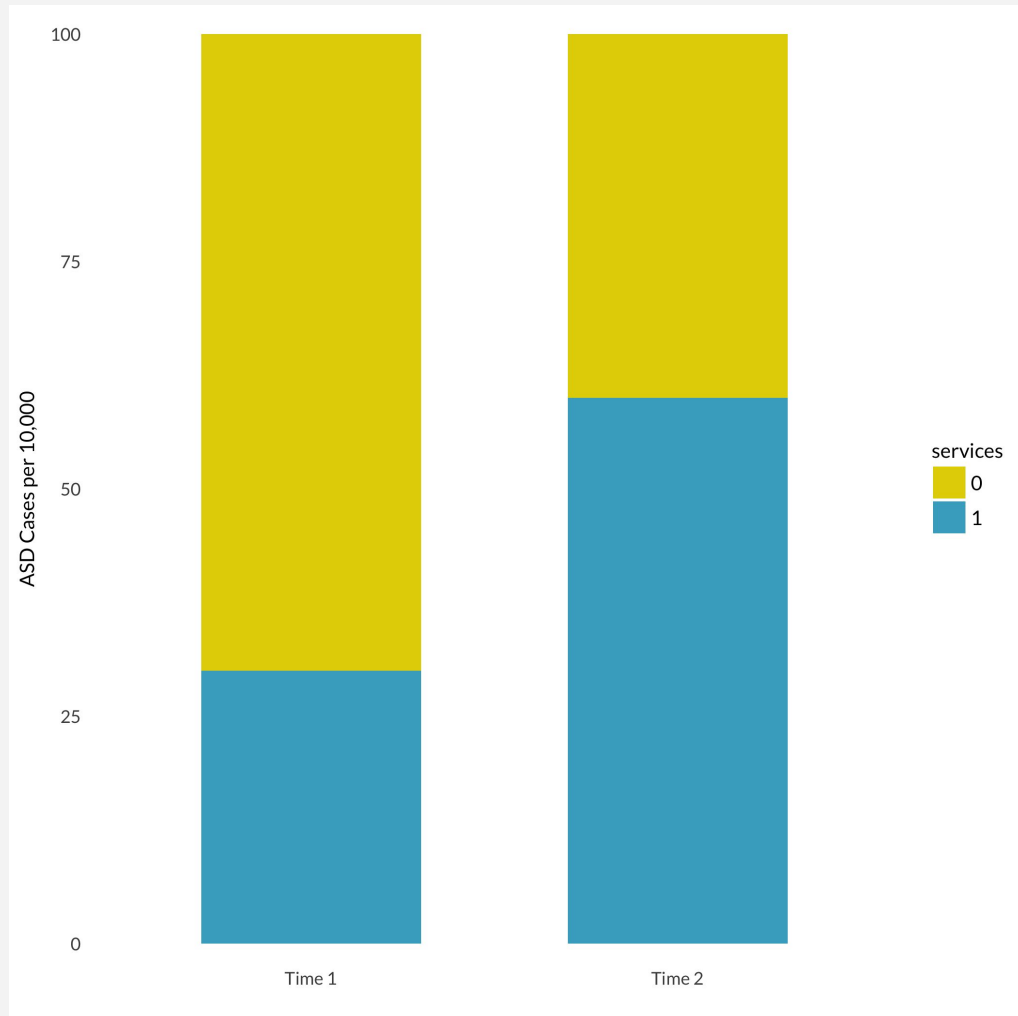
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



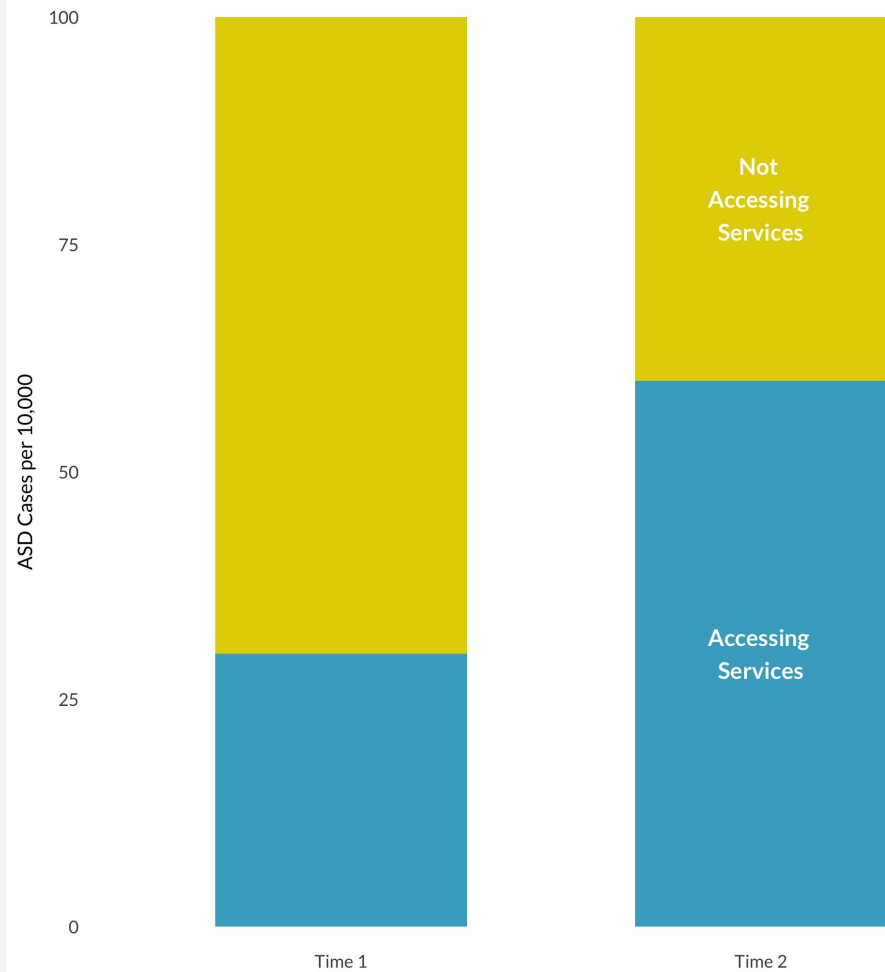
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



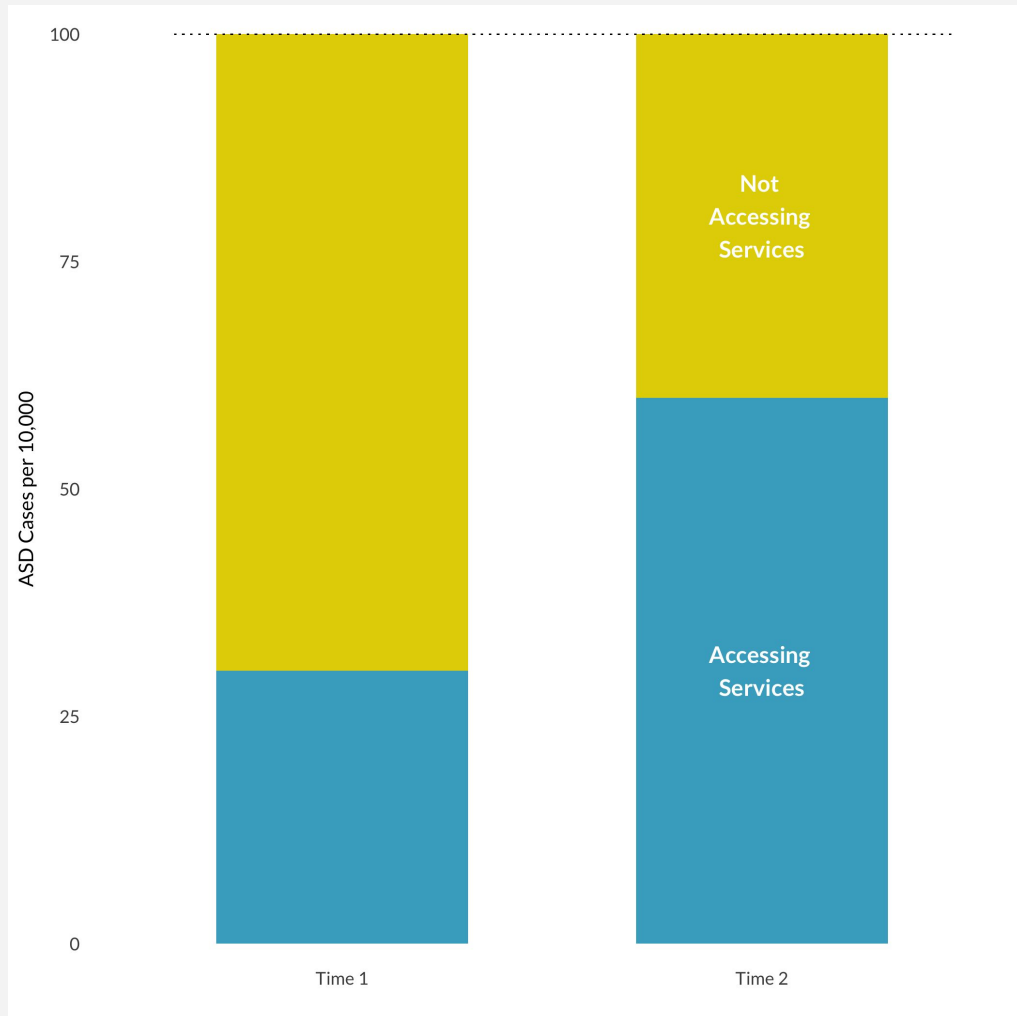
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



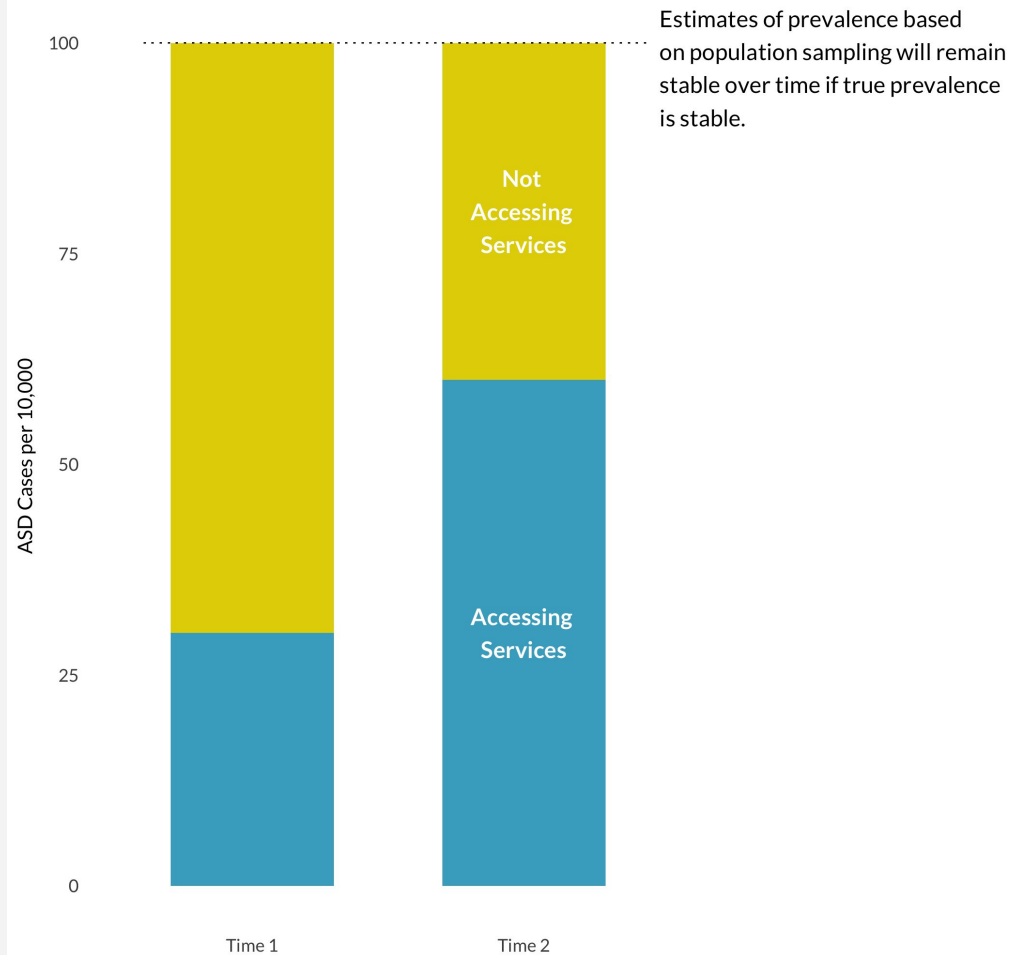
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



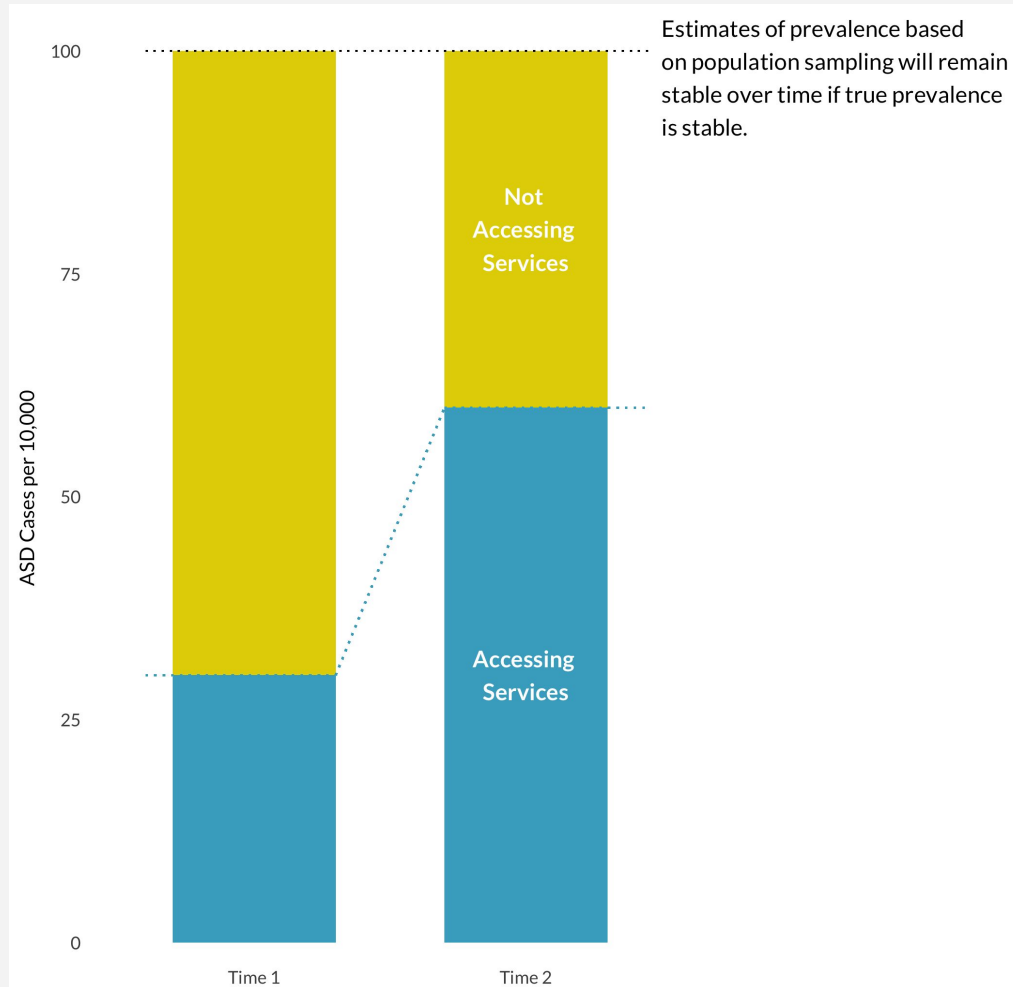
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



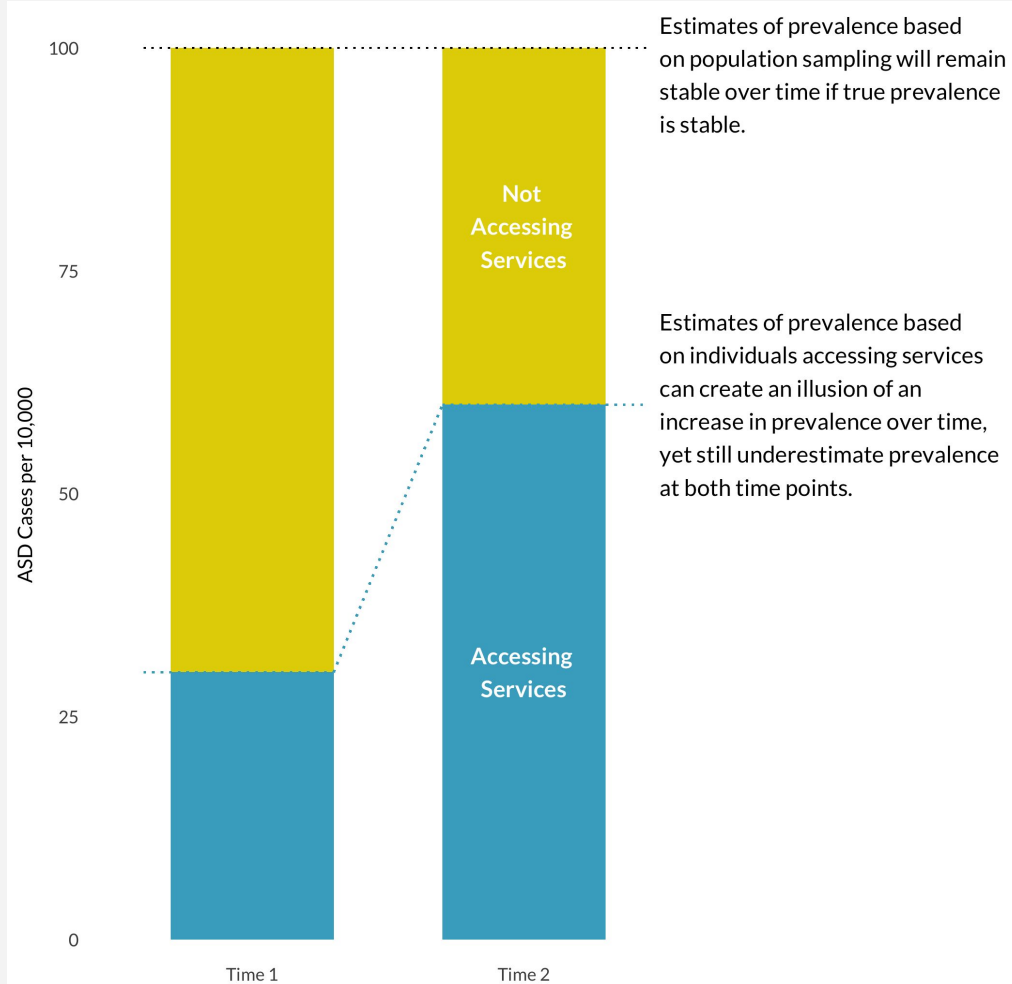
Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration

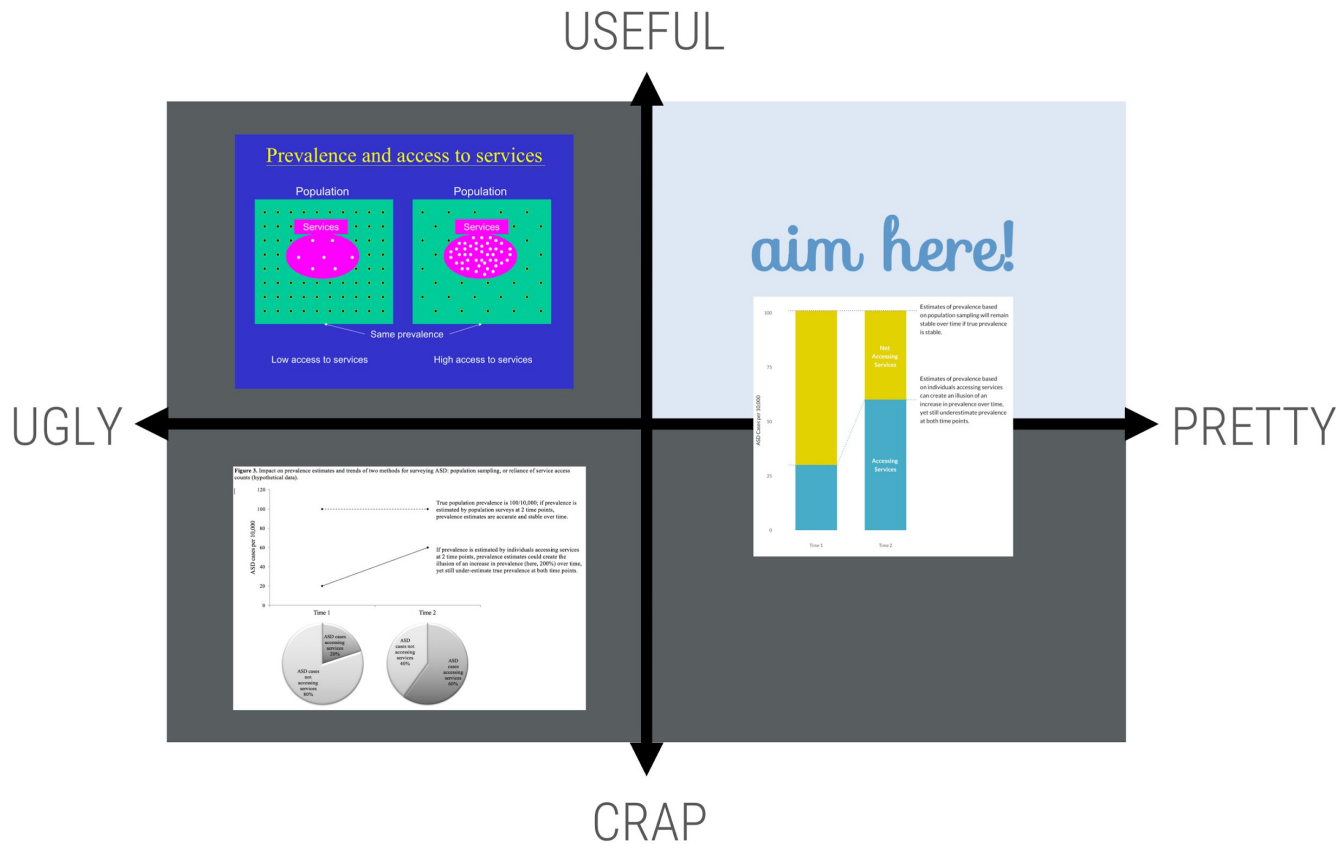


Source: <https://apreshill.github.io/ohsu-biodatavis>

Iteration



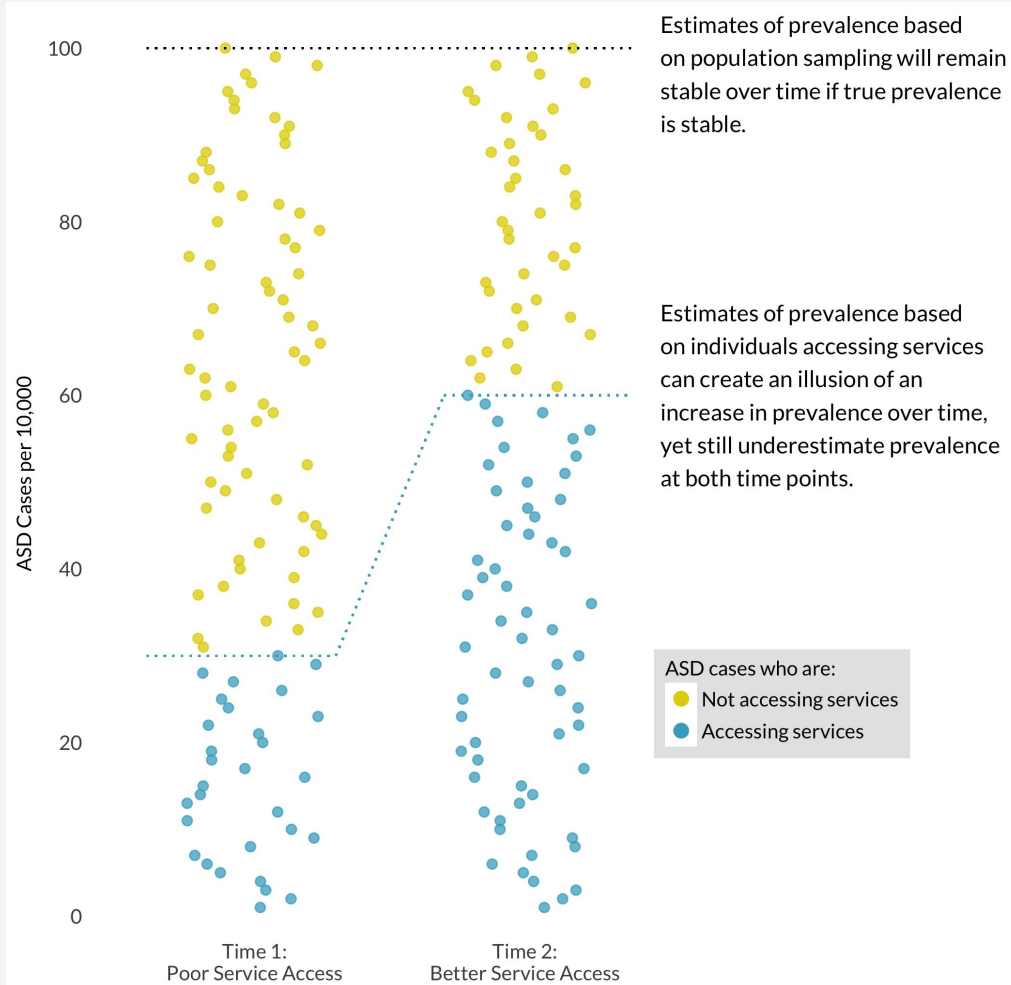
Activity



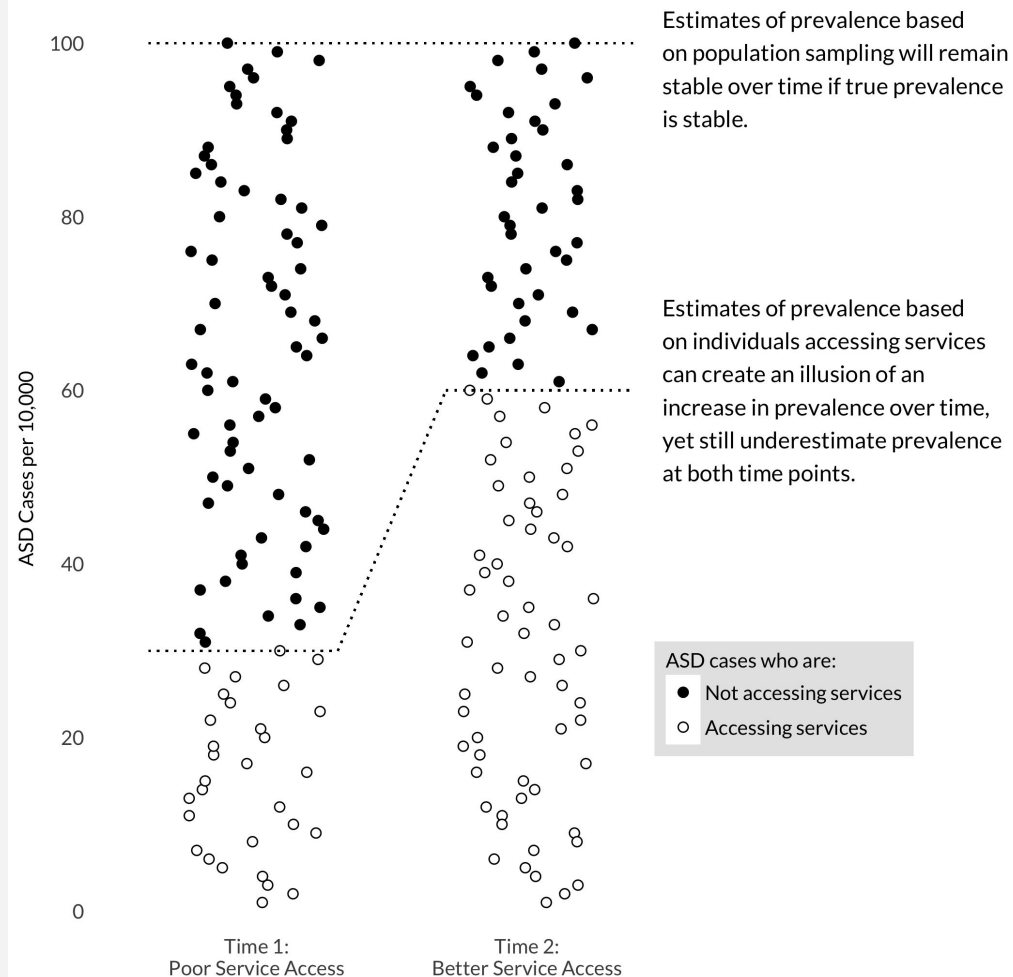
Source: Jackie Wirz

Source: <https://apreshill.github.io/ohsu-biodatavis>

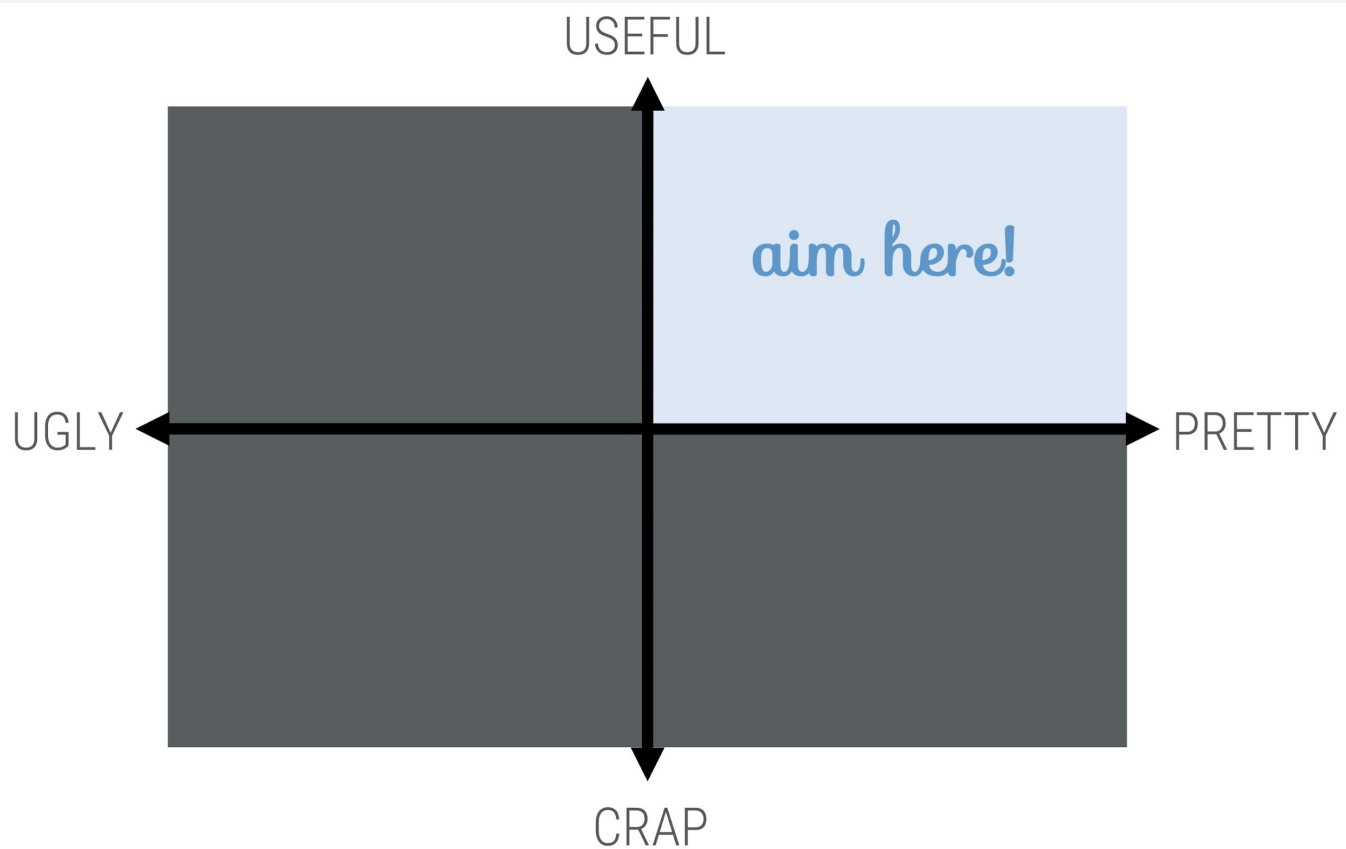
Iteration



Iteration



Your turn: Take a sad plot & make it better

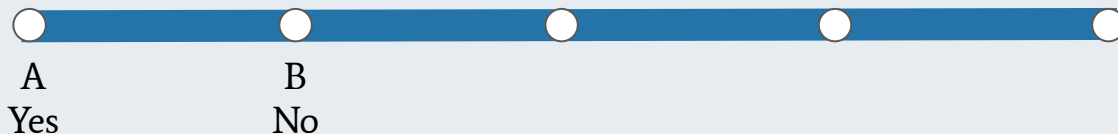


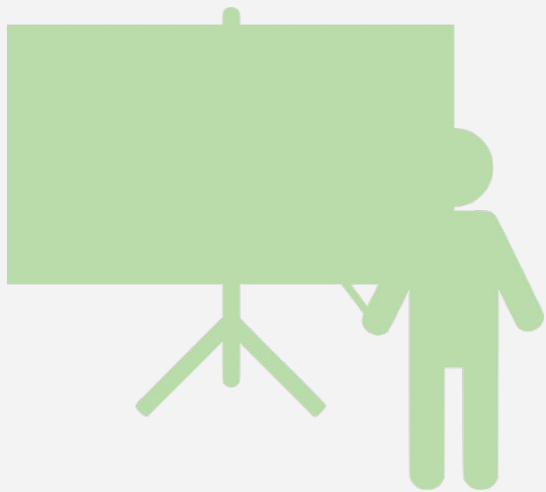
Source: Jackie Wirz

Final Project Video



Q3: Should you spend time making your final report viz pretty, useful, and consistent?

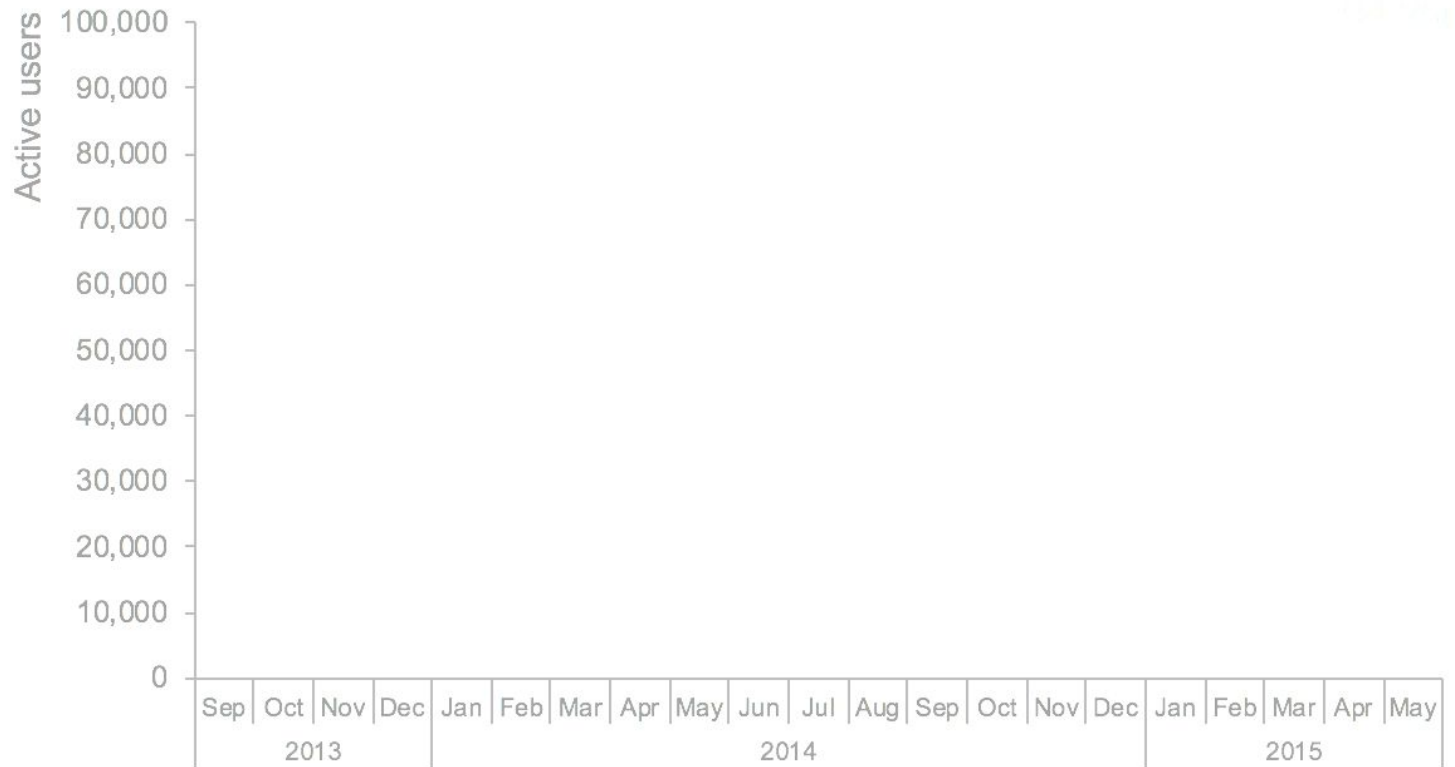




Presentations: for listening

- don't read directly off slides
- use animation to build your story (not to distract)
- introduce your axes
- benefit: words to explain out loud what you're showing

Moonville: active users over time



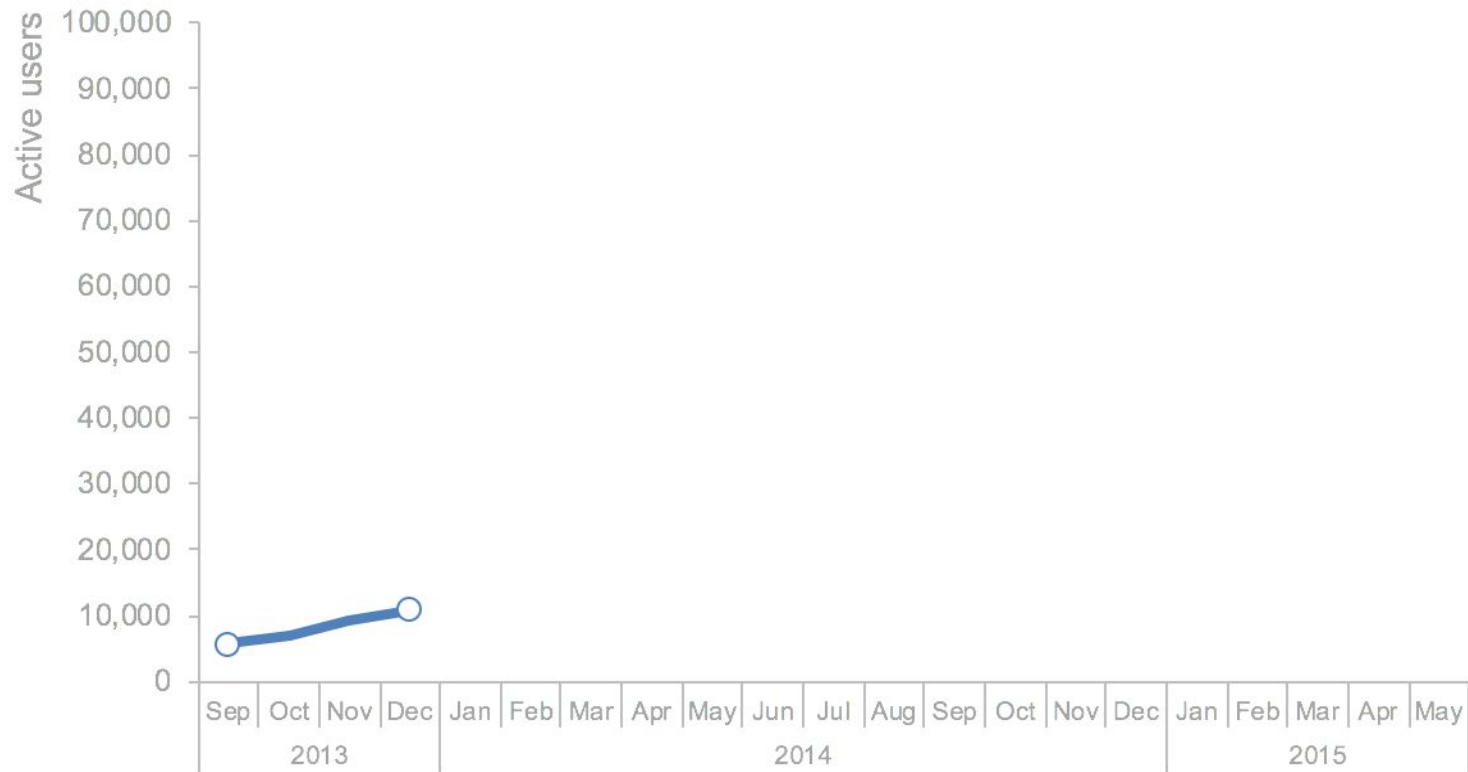
Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Moonville: active users over time



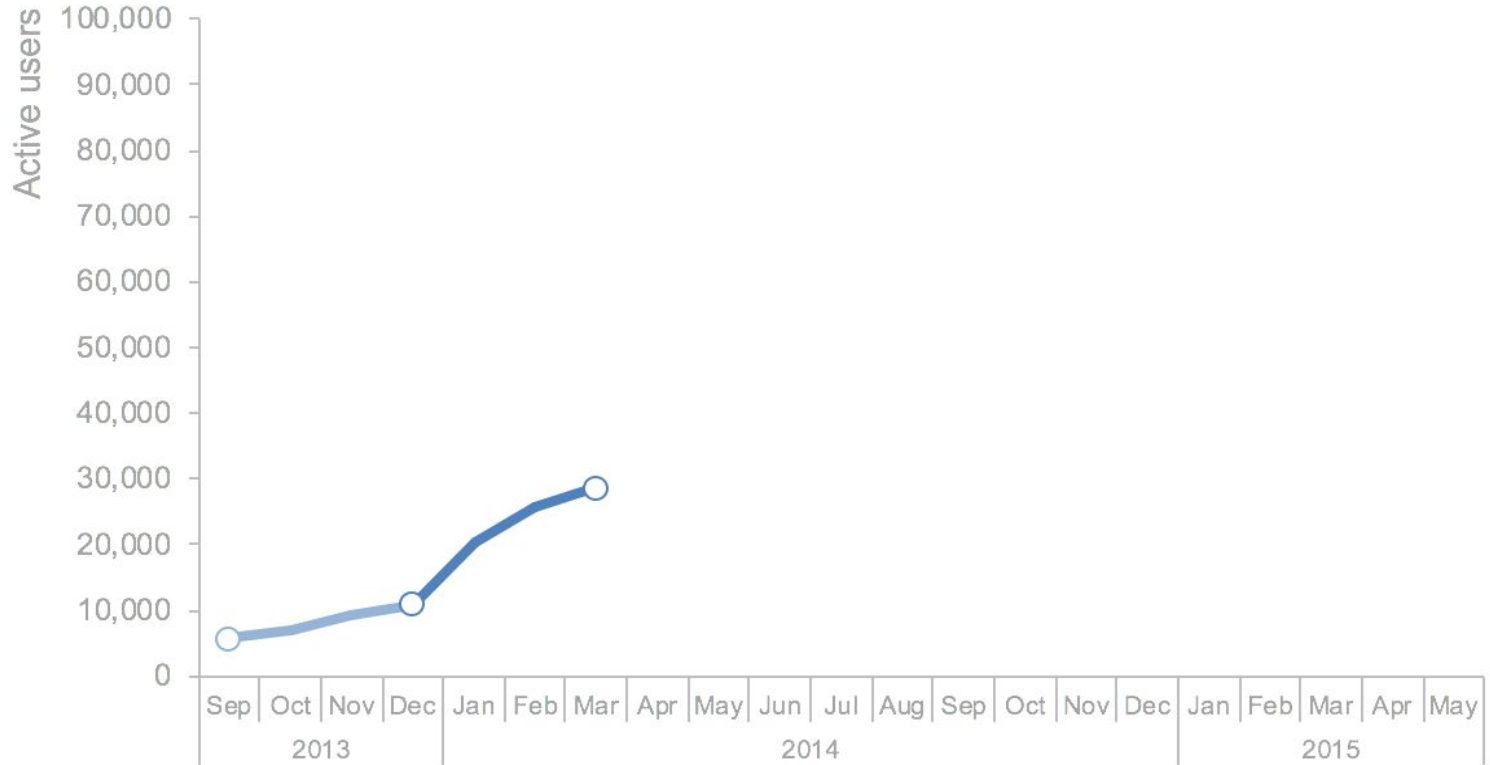
Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Moonville: active users over time



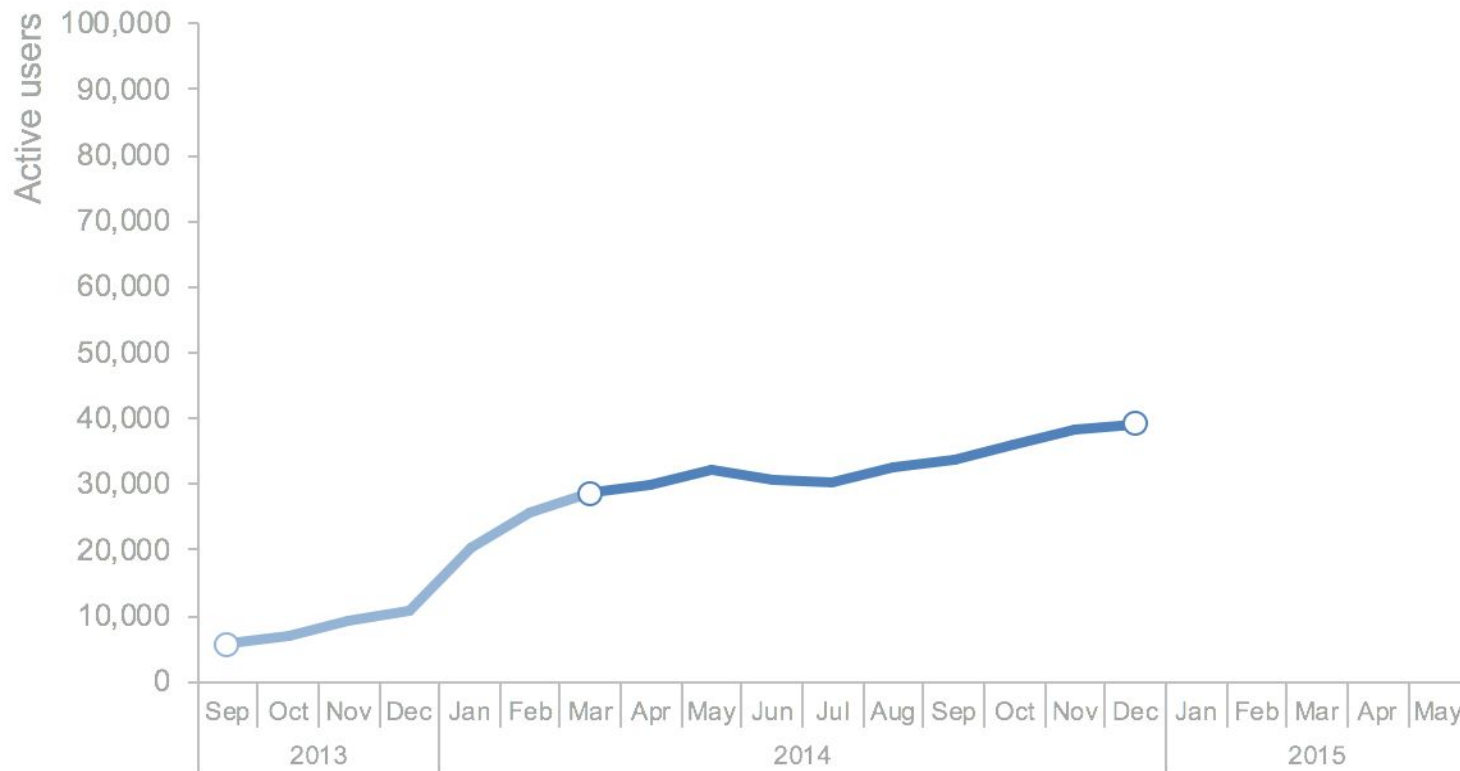
Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Moonville: active users over time



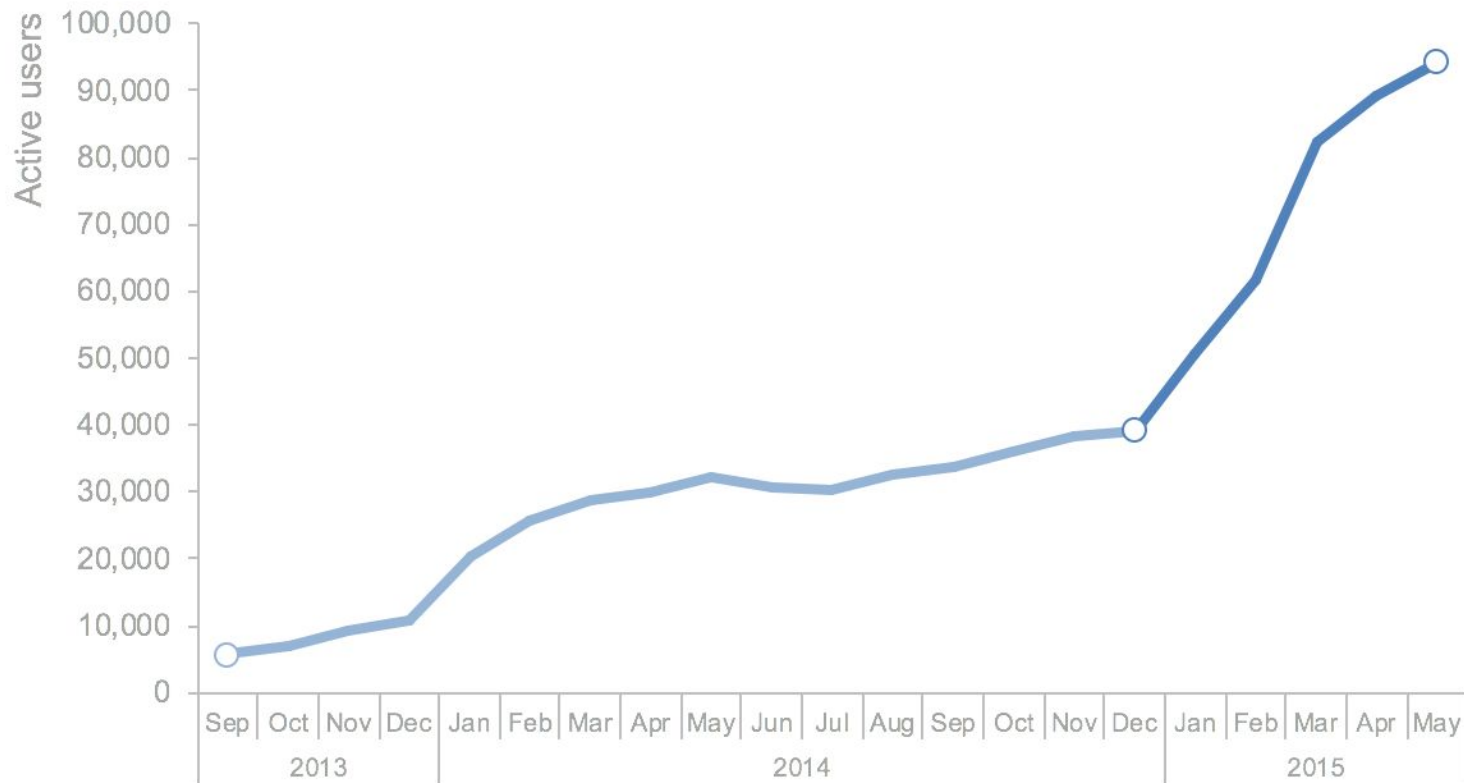
Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Moonville: active users over time

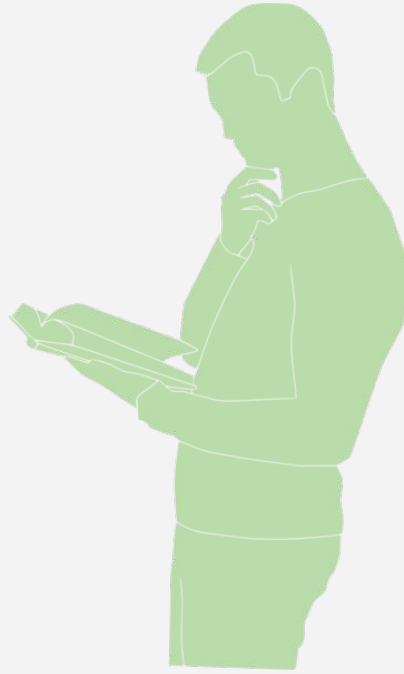


Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Moonville: active users over time



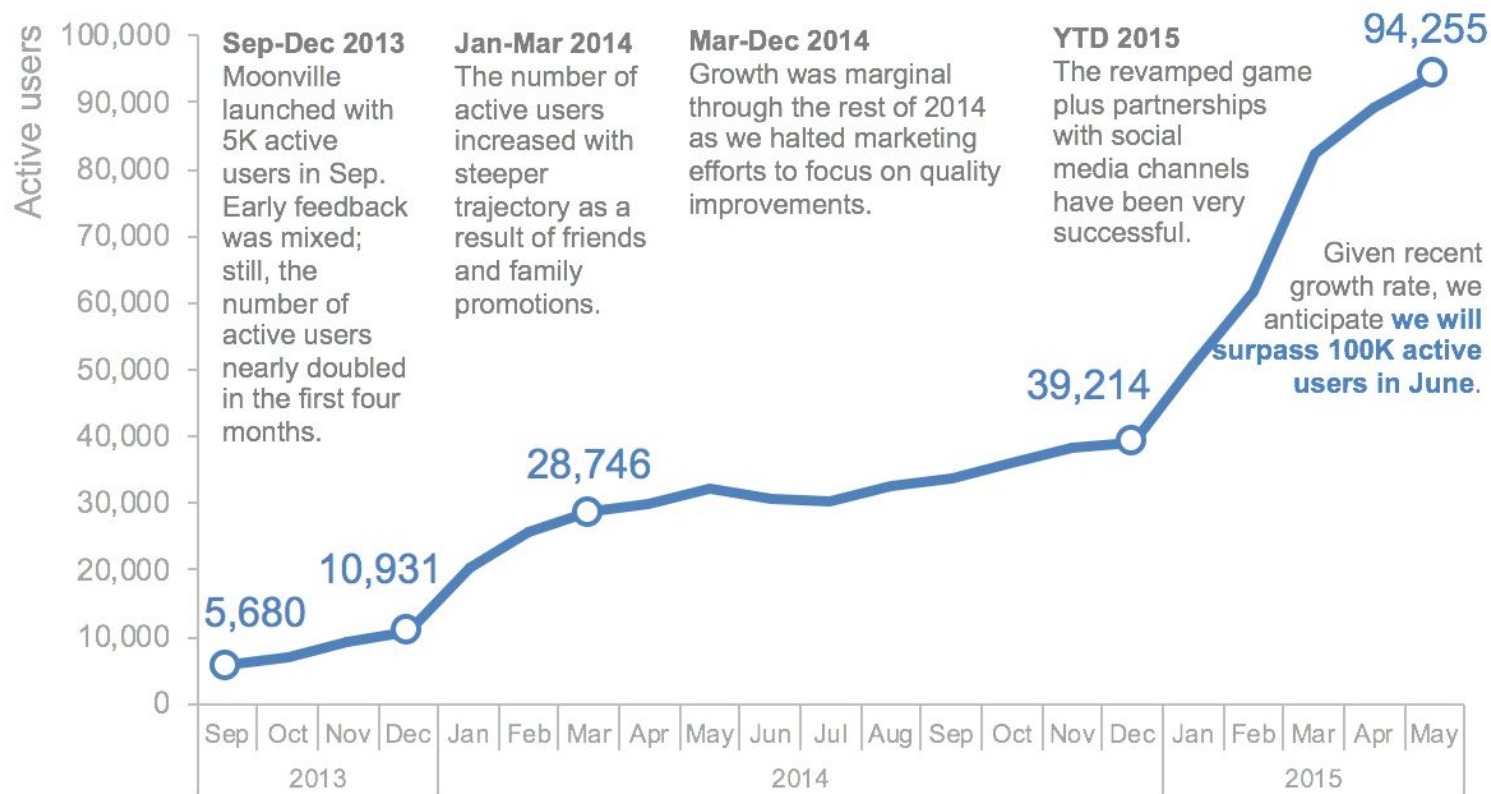
Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.



Reports: for reading

- more on a single visualization
- explanation must be there in text
- Benefit: people have time to look at what you've sent

Moonville: active users over time

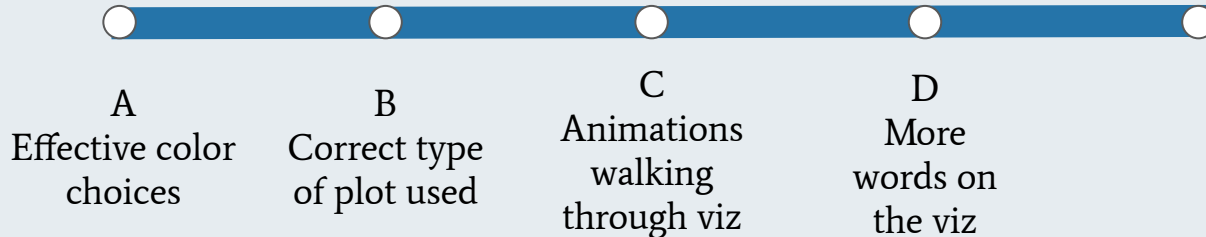


Data source: ABC Report. For purpose of analysis "active user" is defined as the number of unique users in the past 30 days.

Reports vs. Presentations



Q4: What would you expect of a visualization in a report (but not on an oral presentation)?

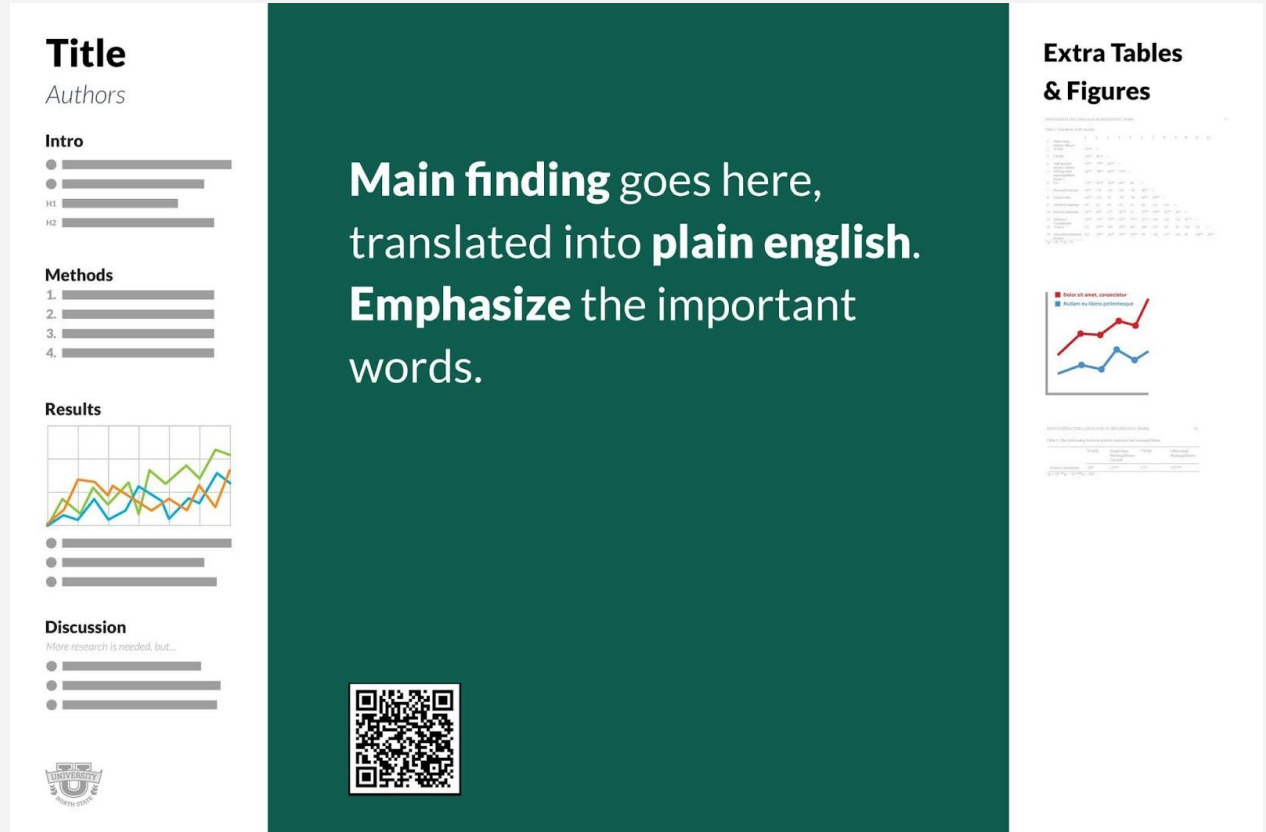


Figures

- ☐ Have informative titles
- ☐ Appropriate figures used
- ☐ Adhering to 'less is more'
- ☐ Mappings clear (maybe a legend)
- ☐ Have consistent colors
- ☐ Take-home message clear
- ☐ Alignment all good
- ☐ Appropriate whitespace

Note: This **takes longer than you expect** it would. You can spend hours on a single figure. But, you worked really hard to understand the data and do the project. Now it's time to show it off! You get to do this through your presentation of your results - **spend the time to do this effectively**.

We don't need it for this class, but you might in the future



Better poster template: <https://osf.io/6ua4k/>